1. Find integration of $f(x)$ with respect to x using Trapezoidal rule with 4 decimal points if

i	a	b	$f(a)$	$f(b)$
1	3.40	4.17	43.601	73.211
2	3.60	4.79	44.003	78.971
3	3.70	5.09	44.010	78.004
4	3.77	5.11	45.000	79.342
5	3.80	6.12	46.708	78.994
6	3.81	6.34	50.000	79.005
7	3.88	6.50	51.230	78.678
8	4.01	6.64	53.003	79.567

2. Find integration of $f(x)$ with respect to x using Trapezoidal rule with 4 decimal points if

i	a	b	$f(a)$	$f(b)$
1	3.0046	5.0064	-245.1	-145.7
2	3.0055	5.0089	-244.0	-143.2
3	3.0100	5.0099	-240.0	-142.5
4	3.0102	5.0113	-231.9	-142.3
5	3.0122	5.0128	-221.1	-142.0
6	3.0125	5.0345	-217.7	-141.0
7	3.0127	5.0500	-217.6	-140.7
8	3.0130	5.0642	-218.2	-140.1
9	3.0134	5.0788	-220.6	-138.0
10	3.0141	5.0853	-224.3	-137.6

3. Given the function $f(x) = 65x^{log_7(0.04x)} + \sqrt[x]{0.5x^2}$. Find integration of $f(x)$ with respect to x using Trapezoidal rule with 5 decimal points if

i	a	b
1	3.4021	5.0201
2	3.6004	5.0206
3	3.7009	5.0210
4	3.7703	5.0217
5	3.8032	5.0219
6	3.8122	5.0222
7	3.8812	5.0223
8	4.0109	5.0225
9	4.0222	5.0228
10	4.0239	5.0230
11	4.0240	5.0239
12	4.0255	5.0244

Assoc. Prof. Dr. Noor Fadiya Mohd Noor

4. Use Trapezoidal rule to estimate the first integration of $f(x_i) = 5\sec^3(5x_i)\,(4x_i - 1)^{-\frac{1}{3}}$ for $i = 12,..,45$ in 7 decimal points if $x_{12} = 0.03$ and $x_{i+1} - x_i = 0.00005$.

5. Use Trapezoidal rule to estimate the first integration of $f(x_i) = \frac{3{x_i}^{\pi}}{\sqrt{-4ln(1.6^{-2x_i})}}$ for $i = 83,..,100$ in 9 decimal points if $x_{83} = 0.03$ and $x_{i+1} = x_i + 0.002$.

6. Find integration of $f(x)$ with respect to x using Simpson 1/3 rule with 4 decimal points if

i	a	b	$f(a)$	$f\left(\frac{a+b}{2}\right)$	$f(b)$
1	0.06	1.23	35.5	46.7	60.1
2	0.08	1.45	37.8	46.9	63.3
3	0.10	1.46	39.0	47.2	64.5
4	0.11	1.47	38.9	47.8	65.7
5	0.14	1.48	41.1	47.8	66.0
6	0.15	1.50	41.0	47.9	67.1
7	0.18	1.51	39.8	48.0	68.1
8	0.20	1.52	39.9	48.3	68.9
9	0.21	1.54	39.8	48.5	69.2

7. Find integration of $f(x)$ with respect to x using Simpson 1/3 rule with 4 decimal points if

i	a	b	$f(a)$	$f\left(\frac{a+b}{2}\right)$	$f(b)$
1	-45.7	-25.1	123	176	234
2	-45.0	-24.6	124	179	235
3	-44.5	-24.3	126	181	256
4	-44.3	-22.0	127	188	267
5	-40.1	-21.8	130	189	279
6	-39.4	-21.3	133	190	288
7	-39.3	-20.0	145	192	296
8	-39.1	-19.7	156	193	301
9	-38.1	-19.6	167	196	312
10	-37.6	-19.5	177	199	323
11	-37.0	-19.4	190	201	378
12	-36.6	-19.2	210	204	381
13	-36.3	-19.0	223	206	389

Assoc. Prof. Dr. Noor Fadiya Mohd Noor

8. Given the function $f(x) = \frac{-2}{\lg(4\sin(5x)+\sqrt{x})} + 4x^{-e^{-x}}$. Find integration of $f(x)$ with respect to x using Simpson 1/3 rule with 6 decimal points if

i	a	b
1	1007.432	2034.123
2	1007.640	2034.322
3	1007.999	2034.422
4	1008.007	2034.472
5	1008.023	2034.478
6	1008.056	2034.489
7	1008.096	2034.499
8	1009.001	2035.502
9	1009.012	2035.506
10	1009.014	2035.509
11	1009.023	2035.511
12	1009.032	2035.514
13	1009.035	2035.519
14	1009.040	2035.522
15	1009.048	2035.524
16	1009.055	2035.529
17	1009.087	2035.535
18	1009.100	2035.540

9. Use Simpson 1/3 rule to estimate the first integration of $f(x_i) = -5\left(\sqrt[4]{2x+7}\right)sinh^{-1}\left(e^{40\cos(5x^2)}\right)$ for $i = 41,..,100042$ in 7 decimal points if $x_{41} = 0.00005$ and $x_{i+1} = x_i + 0.00005$.

10. Use Simpson 1/3 rule to estimate the first integration of $f(x_i) = \frac{7x+\left(\sqrt{\frac{4}{xe^{2x}-\ln x}}\right)^{-x}}{5x^{7\cosh\left(\frac{3x}{2}\right)}}$ for $i = 24,..,95$ in 9 decimal points if $x_{24} = 0.78$ and $x_{i+1} - x_i = 0.001$.

Assoc. Prof. Dr. Noor Fadiya Mohd Noor